MES Apex Arena 2.0

Participant Rulebook and Modules Guide

Module 1: Bridge Breaker (Technical)

Module 2: Code Crackers (Logical)

Module 3: Mech Mayhem Relay (Entertaining and Engaging)

20 teams | 4 members per team | Open to all departments

Date, time and venue: to be announced

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1. Welcome to Apex Arena 2.0

Apex Arena is the flagship event of the **Mechanical Engineering Society (MES)** at The University of Lahore. It is a day of building, thinking and having fun together. Teams from **every department** compete in three different challenges. You do not need any technical knowledge. You only need teamwork, a clear head and a good sense of humour.

This guide tells you everything you need to know: what each module is, how to prepare, how scoring works and what the rules are. Please read it fully with your team before the event day.

What Apex Arena 2.0 is about

- Building something that works, in a short time, with limited materials (**Technical**).
- Solving puzzles with logic and clear thinking (**Logical**).
- Working with your team in a fast and funny relay (**Entertaining and Engaging**).
- Making friends with students from other departments.
- Preparing for bigger engineering competitions through teamwork and good habits.

The three modules in one look

Module	Type	What you do	Marks
1. Bridge Breaker	Technical	Build a bridge from popsicle sticks, paper straws and string. Then it is loaded with weights until it fails.	100
2. Code Crackers	Logical	Solve 5 puzzles. Each answer is one digit. Together they make a 5-digit code.	100
3. Mech Mayhem Relay	Entertaining and Engaging	A 3-station relay: Blind Guide, Straw Blow Race and Balance Walk.	100

Please note. The date, time and venue of Apex Arena 2.0 are not final yet. MES will announce them officially. This guide explains the modules and rules, which will stay the same.

2. Event at a glance

Event	MES Apex Arena 2.0
Organised by	Mechanical Engineering Society (MES), The University of Lahore
Date	To be announced
Time	To be announced
Venue	To be announced
Number of teams	20 teams only (first come, first served)
Team size	4 members per team
Open to	Students from all departments
Modules	3 (Technical, Logical, Entertaining and Engaging)
Total marks	300 (each module is out of 100)
Registration	Details will be announced by MES
Website	mesuol.xo.je

How the day will run (expected order)

The exact times will be shared once the date and venue are final. The order and the approximate length of each part are expected to be:

Part	What happens	Approximate time
Registration and opening briefing	Teams check in, ID is checked, rules are explained	About 20 minutes
Module 1: Bridge Breaker	Briefing, planning, building and load testing	About 70 minutes
Break	Rest, water and tidy-up	About 15 minutes
Module 2: Code Crackers	Briefing and puzzle solving	About 50 minutes
Module 3: Mech Mayhem Relay	Heats of 4 teams at a time	About 60 minutes
Score tally and refreshments	Scores are added up	About 30 minutes
Prize distribution and closing	Results, awards and photos	About 40 minutes

These timings are a guide. MES may change them on the day.

3. Who can take part, teams and registration

Who can take part

- Any currently enrolled student of The University of Lahore, from any department or semester.
- No technical knowledge or experience is needed for any module.
- Teams with members from different departments are welcome and encouraged.
- Mixed teams with students from other universities are not allowed unless MES announces otherwise.

Teams

- Every team has **exactly 4 members**.
- Choose one **Team Lead**. The Team Lead is the only person who speaks to judges during a run or test.
- A student can be in **only one team**.
- Choose a respectful team name. Offensive or misleading names will be rejected.
- All 4 members must be present at the start of every module. If a member is missing, tell the organisers early. A team may compete with fewer members only if the organising committee approves it before the module begins.
- Team members cannot be changed after registration closes, except with written approval from the organisers.

Registration

- Only **20 teams** will be accepted. Places are given in the order of registration.
- Register through the official MES channel announced by MES. Registration details, including the last date, will be shared with the event announcement.
- Give correct names and details. False information can lead to disqualification.
- Keep your registration confirmation. You may be asked to show it at check-in.

Identification

- Every participant must carry a valid **University of Lahore student ID card**.
- A CNIC or other valid identity card may also be asked for.

4. Before and on the event day

How to prepare with your team

- Read this guide together. Make sure everyone knows the rules of all three modules.
- Decide roles inside the team (see the tips in each module).
- Try the practice puzzles in Section 14. They show you the style of Code Crackers.
- Sleep well, eat before you come and drink water.

What to bring

- Your University of Lahore student ID card.
- A pen or pencil for each member (spare ones are useful).
- A water bottle.
- Comfortable shoes (you will stand, walk and run around).

What MES provides

- All building materials for Module 1 (sticks, straws, string, tape, scissors, ruler and pencil).
- All puzzle papers for Module 2.
- All relay equipment for Module 3 (cups, ping-pong balls, straws, books, cones and blindfolds).
- Judges, timekeepers and volunteers.

What not to bring

- Your own building materials, tools or pre-made parts.
- Books, notes, calculators or smart watches for Modules 1 and 2.
- Anything dangerous. Weapons and dangerous items are not allowed.

Reporting and attendance

- Arrive at the time announced and report to the registration desk.
- Be in your seat or at your lane before each module starts. Late teams may lose time or be disqualified from that module.
- Listen carefully to every announcement. Official announcements on the day come first.

Mobile phones

Phones must be switched off or on silent and kept in your bag during Modules 1 and 2. During Module 3, keep them in your bag and away from the lanes.

5. How scoring works

Each module is worth **100 marks**. The total for all three modules is **300 marks**. The team with the most marks wins.

Module	Where the 100 marks come from
1. Bridge Breaker	Load held (60) + Strength-to-weight ratio (20) + Neatness (10) + Creativity (10)
2. Code Crackers	Correct digits (5 x 16 = 80) + Speed bonus (20), minus 4 for each hint used
3. Mech Mayhem Relay	Three stations (3 x 25 = 75) + Speed bonus (25)

Good to know

- Scores are written on an official scoresheet. The judge and the Team Lead sign it.
- Judges use only the criteria in this guide. You can ask a judge to explain how a mark was given.
- The scores of the three modules are added together after Module 3.
- If two teams have the same total, the tie-breaker rules in Section 9 are used.

6. Module 1: Bridge Breaker (Technical)

Your team designs and builds a bridge from basic materials in 40 minutes. Then the judges load your bridge with weights until it fails. You score for how much it holds, how light and neat it is and how creative your design is.

Category	Technical
Team	4 members
Marks	100
Time	5 minutes planning (sketching only) + 40 minutes building + 2 minutes test per team
Goal	Build a bridge that crosses the test gap and holds as much weight as possible, while being light, neat and creative.

What you get (same kit for every team)

- 60 popsicle sticks, 30 paper straws, 3 metres of string and 2 metres of masking tape
- 1 pair of safety scissors (shared by the team), 1 ruler and 1 pencil
- No other materials, tools or pre-made parts are allowed. You do not have to use everything.

Bridge size and test setup

- The bridge rests on two supports with a **clear gap of 30 cm** between them.
- Length: **40 cm to 50 cm** (so it sits at least 5 cm on each support). Width: at least 6 cm. Height: no more than 25 cm.
- Build a **string loop** at the centre of the underside of your bridge. The judges hang a bucket from this loop.
- Weights are added to the bucket in steps of 500 g until the bridge fails or the limit of 10 kg is reached.

How the round works

1. **Briefing and kits (5 min).** Kits stay closed until the judge says start.
2. **Planning (5 min).** Talk and sketch your design on paper. You may not touch the materials yet.
3. **Building (40 min).** You get a warning at 10 minutes and at 2 minutes left. At the end signal, all hands off the bridge.
4. **Weigh-in.** A judge weighs your bridge and checks its size.
5. **Load test (2 min per team).** The team places the bridge on the test supports. Weights are added step by step with about 5 seconds between steps.
6. **End of test.** The test ends when the bridge fails or reaches 10 kg. The last full load held for 5 seconds is your load.

Your bridge fails when:

- Any part breaks or collapses.
- It slips off a support.
- The bucket sags more than 8 cm below the deck.
- The bucket touches the table or floor.

Scoring (100 marks)

What is judged	Marks	How it is marked
Load held	60	6 marks for every full 1 kg held, up to 10 kg
Strength-to-weight ratio	20	Load held divided by the weight of your bridge. The best team gets 20 and others get a share in proportion.

What is judged	Marks	How it is marked
Neatness	10	Clean joints, straight lines, tidy finish. Two judges give a mark and it is averaged.
Creativity	10	Original and clever design. Two judges give a mark and it is averaged.

Example: a bridge that holds 7 kg gets 42 marks for load (7 x 6). A lighter bridge that holds the same load gets a higher strength-to-weight ratio, so it scores more of the 20 marks.

Rules

1. Use only the materials in your kit.
2. No pre-built parts. Start building only after the start signal.
3. Join parts **only** with the supplied string and masking tape. Glue, staples, nails and other fasteners are not allowed.
4. You may cut, bend, notch, roll and tie sticks and straws. You may not burn, melt or soak them.
5. Your bridge must fit the size limits. If it does not fit, it is not tested until you fix it. If you cannot fix it within the build time, 20 marks are taken from the load score.
6. You must build the string loop yourselves, at the centre of the underside.
7. All 4 members take part in building. Only team members may touch the bridge.
8. After the end signal, do not touch the bridge, except to carry it to the weigh-in when a judge tells you to.
9. At the test station only the Team Lead and one other member place the bridge. The others stay behind the line.
10. Do not copy designs from other teams and do not look at their kits.
11. No phones and no internet searches during the build.
12. Judges' measurements and decisions are final.

Safety in Module 1

- Carry scissors with the points down. Never wave them.
- Keep hands and feet away from the bucket during loading.
- Everyone except the testing pair stays behind the safety line.
- Tell a volunteer at once if someone gets cut or hurt.

You may be disqualified from this module for:

- Using outside materials or banned fasteners
- Starting early
- Damaging another team's bridge
- Ignoring safety instructions
- Unsporting behaviour

Tie-breaker

If two teams have the same score, the team with the higher strength-to-weight ratio ranks higher. If still tied, the team with the higher neatness score ranks higher. If still tied, the team with the lighter bridge ranks higher.

Tips for a strong bridge

- **Use triangles.** A triangle does not bend easily. A flat rectangle bends.
- **Share the jobs.** For example: one designer, one cutter, one tier of joints, one checker who tests joints and measures.
- **Make the middle strong.** The weight hangs at the centre, so strengthen the middle.
- **Pull the joints tight.** A loose string joint is where bridges fail first.
- **Do not overbuild.** A very heavy bridge loses marks in the strength-to-weight ratio.
- **Check the size early.** Measure the length, width and height before time runs out, and keep a few minutes for fixing.
- **Make it neat.** Cut ends straight and tuck loose string. Neatness is 10 easy marks.

7. Module 2: Code Crackers (Logical)

Code Crackers is a puzzle challenge on paper. Each team gets a sealed envelope with **5 puzzles**. Every puzzle you solve gives you **one digit**. The five digits, in the right order, make a **5-digit code** that you hand to a judge. Think clearly, share the work and check each other's answers.

Category	Logical
Team	4 members
Marks	100
Time	5 minutes briefing + 45 minutes solving
Materials	Provided by MES (paper only)
Goal	Solve all five puzzles, put the digits in the right order and hand in the correct 5-digit code before time runs out.

What is in your envelope

- 5 puzzle cards (Puzzle 1 to Puzzle 5) with your Set number on them
- 1 answer sheet with five digit boxes
- Rough paper and a pencil

Every team gets a different set. No two teams have the same puzzles or the same final code, but all sets are equally hard. So there is nothing to gain by looking at another team's papers.

The five puzzles

#	Puzzle	What you do	The digit is...
1	Cipher	A number word is written in a secret code. Move each letter back by the shift number.	The number named by the decoded word (for example SEVEN gives 7)
2	Number pattern	Find the rule in a row of 5 numbers and work out the next number.	The last digit of the next number
3	Riddle	Use 4 clues to find a hidden single-digit number.	The hidden number
4	Matchstick	Move exactly one matchstick from one number to a different number to make a wrong equation true.	The number on the right of the = sign after fixing
5	Logic	Four friends each hold a different number card. Use 3 clues to find who holds which card.	The card held by the person named in the question

How the round works

1. **Briefing (5 min).** The judges explain the rules. Envelopes stay sealed.
2. **Start signal.** Open your envelope. The clock starts for every team together.
3. **Solving (45 min).** Split the puzzles among you if you like. Write the digits in the order Puzzle 1 to Puzzle 5.
4. **Write each digit** in its box on the answer sheet when you are sure of it.
5. **Hand in.** The Team Lead raises a hand. A judge checks the answer sheet and notes the time.
6. **End.** At the 45-minute signal, all pencils down. Answer sheets are collected.

Hints

You may ask a judge for up to **2 hints**. Tell the judge which puzzle you need help with. A hint gives a small clue for that puzzle and **costs 4 marks**. Hints are optional. An unsolved puzzle costs you 16 marks, so a hint can be worth it.

Scoring (100 marks)

What is marked	Marks	How it is marked
Correct digits	80	16 marks for each correct digit in the correct place (5 puzzles)
Speed bonus	20	Only for teams that hand in the full correct code in time
Hints	minus 4 each	Taken off for every hint used

Order of finishing (full correct code)	Speed bonus
1st, 2nd and 3rd	20, 18 and 16 marks
4th to 10th	12 marks
11th to 20th (within time)	8 marks

Rules

1. Each team works at its own table. Do not look at, copy or talk about answers with other teams.
2. Keep your envelope sealed until the start signal.
3. Use only the materials provided. Phones, smart watches, calculators and books are not allowed.
4. All 4 members stay together in the room. You may leave only for an emergency approved by a judge.
5. Write on the rough paper and the answer sheet only. Do not tear puzzle cards or write on them unless a card says you can.
6. The final code must be written on the answer sheet and handed to a judge. A code spoken to a judge does not count.
7. You may hand in your code up to **3 times**. After a wrong try, the judge only says "incorrect" and does not say which digit is wrong. Each wrong try after the first costs 2 marks from your speed bonus.
8. The speed bonus is decided by the time of your first fully correct hand-in.
9. Keep your voices low. Repeated noise brings a warning and then a 4-mark deduction.
10. Judges cannot explain or confirm answers. They give no help except hints.
11. Judges' decisions are final.

You may be disqualified from this module for:

- Cheating or looking at another team's papers
- Using a phone or other banned device
- Opening the envelope early
- Hiding or tearing puzzle cards
- Unsporting behaviour

Tie-breaker

If two teams have the same score, the team with the faster full-code time ranks higher. If still tied, the team that used fewer hints ranks higher. If still tied, there is a 1-minute rapid riddle round between the tied teams.

Tips for Code Crackers

- **Read all five puzzles first** (2 or 3 minutes). Give each member the puzzle they like best.
- **Write your working.** Cipher: write the alphabet numbers. Pattern: write the gaps. Riddle: write 0 to 9 and cross out numbers.
- **Check each other's answers.** A wrong digit loses 16 marks. A second pair of eyes is worth the time.
- **Do not get stuck.** If a puzzle takes more than about 8 minutes, move on and come back.
- **Use hints wisely.** One hint costs 4 marks, but an unsolved puzzle costs 16.
- **Hand in only when you are sure.** Wrong hand-ins cost speed bonus marks.
- See Section 14 for practice puzzles that show how each type works.

8. Module 3: Mech Mayhem Relay (Entertaining and Engaging)

Mech Mayhem Relay is a fast and funny relay with **three mini-stations**. Three team members each play one station, while the rest of the team guides and cheers. It is easy to watch, easy to enjoy and needs only coordination and teamwork.

Category	Entertaining and Engaging
Team	4 members (3 players and 1 support member)
Marks	100
Time	About 3 minutes per team. Teams run in heats of 4 teams at a time. Each station has a limit of 60 seconds.
Materials	Provided by MES
Goal	Complete all three stations in order, score as many marks as you can at each station and finish the relay in the fastest total time.

Station	Name	Player	What happens
1	Blind Guide	Player 1 (blindfolded)	Stack cups into a pyramid, guided only by your teammates' voices
2	Straw Blow Race	Player 2	Move a ping-pong ball along a lane using only a straw and your breath
3	Balance Walk	Player 3	Walk a short course with a book on your head while carrying a cup of balls

Team roles

- **Players 1, 2 and 3** each play one station. Before the start, the Team Lead writes the player order on the team slip and gives it to the judge. The order cannot change afterwards, except for a medical reason approved by a judge.
- **The support member** (the 4th member) helps guide at Station 1 and cheers the team. They may replace a player who cannot play for a medical reason.
- The team members who are not playing guide the player at Station 1. At the other stations, they may cheer and advise but may not touch anything.

Station 1: Blind Guide

Task. The player is blindfolded and builds a 6-cup pyramid (3 cups at the bottom, 2 in the middle and 1 on top) on a table. The other teammates stand behind a marked line about 2 metres from the table. They guide the player with their voices only, for example "left", "slow" or "two cups forward".

- The player may touch only the cups and the table.
- Teammates stay behind the line. They may not touch the player or the cups.
- A cup that falls can be picked up and used again.
- The blindfold stays on until the judge says "stop".

Scoring (25 marks). 4 marks for each cup correctly placed in the pyramid at the end of the time (up to 24), plus 1 bonus mark if the full pyramid is standing within 45 seconds.

Station 2: Straw Blow Race

Task. The player moves a ping-pong ball along a 2-metre lane from the start line to the finish line using only a drinking straw and their own breath.

- The ball may not be touched, fanned or pushed with anything else. You may not use the straw to flick or push the ball. Blowing only.

- The player may bend down but may not touch the lane or the ball with hands, face or body.
- If the ball leaves the lane, it is put back at the point where it left.
- Teammates may cheer but may not touch the lane, the ball or the player.

Result	Marks
Ball crosses the finish line within 30 seconds	25
Within 45 seconds	20
Within 60 seconds	15
Ball does not reach the finish line	1 mark for each 25 cm covered, up to 7

Station 3: Balance Walk

Task. The player walks a course of about 4 metres. They go around 3 cones and then across the finish line, with a book balanced on their head and a cup holding 3 ping-pong balls in one hand.

- Walk only. Running is not allowed.
- The book may not be held or touched with the hands while walking.
- If the book falls, the player puts it back on the head and continues from where it fell.
- If a ball falls from the cup, it stays on the floor and the player continues.

Section completed	Marks
Cone 1	6
Cone 2	6
Cone 3	6
Finish line	7

Deductions: 2 marks each time the book falls and 1 mark for each ball dropped. The score cannot go below zero.

How the relay runs

1. **Before the start.** The Team Lead writes the player order on the team slip and gives it to the judge.
2. **Heat call.** Four teams are called at a time. Each team stands at the start of its own lane.
3. **Start whistle.** Player 1 begins Station 1 and the relay clock starts.
4. **Handover.** When a station ends (the task is done or 60 seconds are over), the judge calls "stop". The next player begins when the judge says go.
5. **Finish.** The clock stops when Station 3 ends. The judge records your total time and your marks for each station.
6. **Sign-off.** The Team Lead checks the scoresheet and signs it.

Scoring (100 marks)

What is marked	Marks
Station 1: Blind Guide	25
Station 2: Straw Blow Race	25
Station 3: Balance Walk	25
Speed bonus (total relay time, compared across all heats)	25

Order by total relay time (fastest first)	Speed bonus
1st, 2nd and 3rd	25, 22 and 19 marks
4th to 10th	15 marks

Order by total relay time (fastest first)	Speed bonus
11th to 20th	10 marks

Rules

1. Each of the 3 players plays exactly one station, in the order written on the team slip.
2. Follow the judge's whistle and instructions. Starting a station early costs 2 marks.
3. Team members who are not playing stay in the waiting area or behind the guide line. They do not touch the equipment.
4. Do not interfere with other lanes, other teams' equipment or other players.
5. If equipment breaks, the judge pauses the clock and fixes or replaces it. The player starts that station again.
6. A station ends at 60 seconds, even if it is not finished. The marks earned so far are kept.
7. Do not take extra time to rest or discuss tactics between stations.
8. Keep phones and personal things in your bags, away from the lanes. No coaching from the audience.
9. Judges' decisions and recorded times are final.

Safety in Module 3

- Lane floors are kept clear and dry. Volunteers check before every heat.
- Blindfolds are clean cloth bands. The player may take it off at any time and must tell the judge if they feel unwell or uncomfortable.
- Nobody is forced to take a role they are uncomfortable with. Tell the judges at the briefing so roles can be arranged safely.
- No running, pushing or rough behaviour. A volunteer stands next to the Balance Walk to prevent falls.
- Every player gets their own straw. Straws are not shared.

You may be disqualified from this module for:

- Cheating or ignoring a judge
- Deliberately interfering with another team
- Unsafe or rough behaviour
- Offensive behaviour

Tie-breaker

If two teams have the same score, the team with the faster total relay time ranks higher. If the times are also equal, the team with the higher Balance Walk score ranks higher. If still tied, there is a 1-minute rapid riddle round.

Tips for the relay

- **Choose the right player for each station.** Pick calm people for the blindfold, someone with strong breath control for the straw race and someone steady for the balance walk.
- **Practise your voice directions** for Blind Guide. One person should speak at a time, with short and clear words.
- **Be quick at handover.** Be ready at the line so no time is lost between players.
- **Stay calm if something drops.** Pick it up and carry on. Losing 2 marks is better than losing 20 seconds.
- **Have fun.** The audience enjoys cheerful teams.

9. Overall ranking, tie-breakers and awards

Final ranking

The winner is the team with the highest total across the three modules, out of 300 marks.

Module	Marks
1. Bridge Breaker	100
2. Code Crackers	100
3. Mech Mayhem Relay	100
Total	**300**

If teams have the same total

1. A 1-minute rapid riddle round between the tied teams.
2. If that does not decide it, the team with the higher Bridge Breaker score ranks higher.
3. If still tied, the judges decide with a final quick challenge announced on the day.

Awards and recognition

- Certificates for participants who complete the event.
- Shields and prizes for the top teams.
- Special recognition for the best team in each module.
- The final list of prizes will be announced by MES.

Checking your scores

After each module, the Team Lead may check the team's scoresheet. If you think there is a mistake, follow the steps in Section 10.

10. Fair play, judging and disputes

Code of conduct

Everyone must behave respectfully at all times. The following are not allowed:

- Bullying, harassment or discrimination of any kind
- Threats, shouting, abusive or rude language
- Disturbing a module or ceremony
- Damaging university or event property
- Bad behaviour towards judges, volunteers, organisers or other participants

Cheating

The following are treated as cheating:

- Using outside materials, devices or help when not allowed
- Copying or sharing answers, designs or plans with other teams
- Tampering with equipment, scoresheets, puzzle cards or another team's work
- Giving false information at registration
- Any attempt to gain an unfair advantage

A team found cheating in a module may lose all marks in that module or be disqualified from the event, at the judges' decision.

Judging

- Each module has judges chosen by the organising committee.
- Judges use only the criteria written in this guide.
- A judge will not judge a team that includes a family member or close friend.
- You may ask a judge to explain how a mark was given.

Disputes

1. Only the Team Lead may raise a question.
2. It must be about a score, a penalty or a recorded time.
3. Speak politely to the Head Judge of that module within 10 minutes after your run or the end of the module.
4. Judges' and organisers' decisions are final. Arguing with judges or organisers is not accepted.
5. The organising committee may check any suspected rule break.

11. Safety and general rules

Safety and venue rules

- Follow the instructions of volunteers and security at all times.
- Weapons and dangerous materials are not allowed.
- Keep exits and walkways clear.
- Do not damage, move or write on furniture, walls or equipment.
- Do not touch electrical equipment, sound systems or cables.
- Food and drinks only where and when MES announces.
- Smoking and banned substances are not allowed.
- Tell a volunteer at once about any injury, danger or unsafe behaviour.

Violation of these rules may lead to immediate removal from the event.

Photos and videos

MES may take photos and videos during the event for records, social media, reports and future promotion. By taking part, you agree to appear in them. If you do not want to be photographed, please tell the organisers at registration.

Rights of the organisers

The organising committee may:

- Change the schedule, formats or challenges if needed
- Limit registration to the available places
- Disqualify participants who break the rules
- Cancel or postpone the event or a module in special situations
- Update the rules through official announcements
- Decide any unexpected situation in the interest of fair competition

Official announcements

Follow the official MES channels for updates about the date, time, venue and any change in rules. Official announcements always come before anything you hear from other students.

12. Frequently asked questions

Question	Answer
Do I need technical knowledge?	No. All three modules are made for students of every department. Teamwork and clear thinking are enough.
Can my team have members from different departments?	Yes. Mixed teams are welcome.
Can we join with fewer than 4 members?	No. A team has 4 members. If a member is missing, tell the organisers before the module begins. The committee may allow it in special cases.
How many teams will take part?	Only 20 teams. Places are given in the order of registration.
Can we bring our own materials?	No. MES provides all materials, so every team has the same equipment.
Can we use glue in the bridge?	No. Bridge joints use only the string and masking tape that MES provides.
How many hints can we use in Code Crackers?	Up to 2. Each costs 4 marks.
What if our Code Crackers answer is wrong?	You may hand in up to 3 times. After a wrong try, the judge only says it is incorrect. Each wrong try after the first costs 2 marks from the speed bonus.
Do all teams get the same puzzles?	No. Every team gets a different set. The sets are equally hard.
In the relay, what does the 4th member do?	The support member helps guide at Station 1, cheers the team and can replace a player who cannot play for a medical reason.
What if a relay station breaks or malfunctions?	The judge pauses the clock, fixes it and you start that station again.
What if I feel uncomfortable with a role?	Tell the judges at the briefing. Roles can be arranged safely. Nobody is forced.
What if I think a score is wrong?	The Team Lead may politely ask the Head Judge within 10 minutes after the run or the end of the module.
When and where is the event?	The date, time and venue are not final yet. MES will announce them officially.
How will we know about any changes?	Through the official MES announcements.

13. Quick checklist

Before the event

- Team of 4 registered. Team Lead chosen.
- Everyone has read this guide.
- Practice puzzles tried (Section 14).
- Roles decided for each module.
- Relay: players and support member chosen for the three stations.

On the day

- University of Lahore student ID card with you.
- Pen or pencil and water bottle.
- Phones switched off and kept in bags (Modules 1 and 2).
- Reported on time and listened to the briefing.

Remember the key rules

Module	Remember
Bridge Breaker	Kit only. String and masking tape only (no glue). 30 cm gap, bridge 40 to 50 cm long. Strong middle, tight joints, light and neat.
Code Crackers	Own set, own table. No phones or calculators. Up to 2 hints (-4 each). Up to 3 hand-ins. Check every digit before you hand in.
Mech Mayhem Relay	One station per player. 60 seconds per station. Follow the whistle. Be quick at handover. No touching equipment when not playing.

Spirit of the event. Apex Arena is about building, thinking and enjoying together. Win fairly, lose gracefully and help others learn. Good luck to every team!

14. Practice corner (try it yourself)

These practice puzzles are **not** in the real event. They show the style of Code Crackers. Try them with your team first. The answers and full working are on the next pages.

Practice 1: Cipher

Shift = 4. Coded word: **S R I**. Decode it. The digit is the number named by the word.

Practice 2: Number pattern

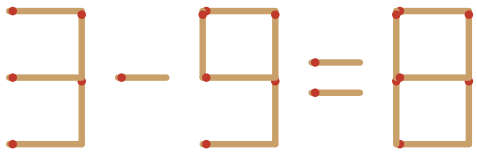
Numbers: **2, 6, 12, 20, 30, ?** Find the next number. The digit is its last digit.

Practice 3: Riddle

I am a single digit (0 to 9). I am an odd number. I am less than 5. If you add 2 to me, the answer is a multiple of 3. I am not a multiple of 5. **What number am I?**

Practice 4: Matchstick

Move exactly one stick from one number to a different number to fix the equation. The digit is the number on the right of the = sign after fixing.



Practice 5: Logic

Four friends, **Bilal, Hira, Mahnoor and Zoya**, each hold a different number card: **4, 6, 7, 8**.

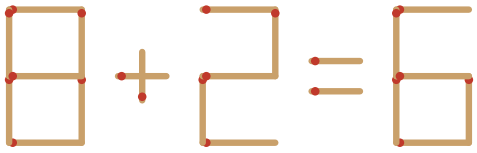
- Bilal and Mahnoor hold cards that add up to 13.
- Bilal holds a card that is 1 more than the card held by Mahnoor.
- Mahnoor holds a bigger card than Zoya.

Question: Which card is Mahnoor holding?

More practice (answers on the last page)

- **Cipher:** shift 6, coded word **T O T K**.
- **Pattern: 3, 5, 10, 18, 29, ?** (the digit is the last digit of the next number)
- **Riddle:** I am a single digit (0 to 9). I am an even number. I am less than 7. If you add 8 to me, the answer is a multiple of 4. I am not a multiple of 5.
- **Logic:** friends Bilal, Daniyal, Mahnoor, Usman with cards 2, 3, 6, 7. Usman holds a bigger card than Bilal. Daniyal and Mahnoor hold cards that add up to 10. Mahnoor holds a card that is 1 more than the card held by Bilal. Which card is Bilal holding?

Matchstick (more practice):



Practice answers and full working

Practice 1: Cipher

Step 1. Understand the code. Every letter of the real word was moved **4 places forward** in the alphabet. To decode, we do the opposite: move every letter **4 places back**. The alphabet strip on the sheet gives each letter a number (A = 1, B = 2 ... Z = 26).

Coded letter	Its number	Minus 4	Result number	Decoded letter
S	19	19 - 4	15	**O**
R	18	18 - 4	14	**N**
I	9	9 - 4	5	**E**

Step 2. Read the word. Putting the decoded letters together gives **ONE**. It is the name of a number.
Step 3. Write it as a digit. ONE = **1**.

Puzzle 1 digit = 1

Practice 2: Number pattern

Step 1. Find the gaps. Subtract each number from the next one.

Pair	2 to 6	6 to 12	12 to 20	20 to 30
Gap	6 - 2 = **4**	12 - 6 = **6**	20 - 12 = **8**	30 - 20 = **10**

Step 2. Look at how the gaps change. The gaps are 4, 6, 8, 10. Each gap is **2 more** than the one before: 4 to 6 (+2), 6 to 8 (+2), 8 to 10 (+2). So the rule is: *the gap grows by 2 every time*.

Step 3. Find the next gap. The last gap is 10, so the next gap is 10 + 2 = **12**.
Step 4. Find the next number. The last number is 30, so the missing number is 30 + 12 = **42**.
Step 5. Take the last digit (units digit). The last digit of 42 is **2**.

Puzzle 2 digit = 2

Practice 3: Riddle

Idea: start with every possible number from 0 to 9. Use the clues **one by one**. Each clue crosses out the numbers that cannot be the answer. What is left at the end is the answer.

Clue 1. "I am a single digit (0 to 9)." We keep all of them: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
Clue 2. "I am an odd number." Odd numbers end in 1, 3, 5, 7, 9. Cross out the others. Left: **{1, 3, 5, 7, 9}**
Clue 3. "I am less than 5." Keep only numbers less than 5. Left: **{1, 3}**

Clue 4. "If you add 2 to me, the answer is a multiple of 3." Test every number that is left:

Number n	n + 2	Is it a multiple of 3?	Keep?
1	1 + 2 = 3	Yes (3 x 1 = 3)	**Keep**
3	3 + 2 = 5	No (5 is not in the 3 times table)	Cross out

Left after Clue 4: **{1}**

Clue 5. "I am not a multiple of 5." Test the numbers that are left:

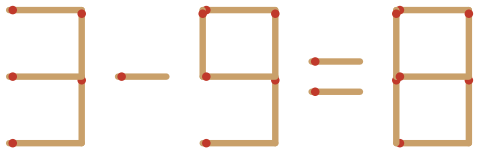
Number n	Is it a multiple of 5?	Keep?
1	No (1 is not in the 5 times table)	**Keep**

Only one number is left: **1**. That is the number in the riddle, so it is the digit.

Puzzle 3 digit = 1

Practice 4: Matchstick

Step 1. Check the equation. The sticks show **3 - 9 = 8**. But 3 - 9 = -6, not 8. So the equation is wrong.



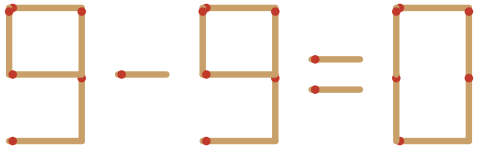
Step 2. Think about what one stick can do. We must take **one stick away** from one number and **add it** to a different number. So one number loses a stick and another number gains a stick. The table shows what each number in this puzzle can become.

Number	Sticks	Remove 1 stick, it can become	Add 1 stick, it can become
left number: **3**	5	nothing	9
middle number: **9**	6	3, 5	8
right number: **8**	7	0, 6, 9	nothing

Step 3. Try every possible move. Choose which number loses a stick and which number gains it. Then check if the new equation is true.

Loses a stick	Gains a stick	New equation	True?
middle	left	9 - 3 = 8	no
middle	left	9 - 5 = 8	no
right	left	9 - 9 = 0	**YES**
right	left	9 - 9 = 6	no
right	left	9 - 9 = 9	no
right	middle	3 - 8 = 0	no
right	middle	3 - 8 = 6	no
right	middle	3 - 8 = 9	no

Step 4. The one that works. Only **9 - 9 = 0** is true. To make it, take the **middle stick** from the **right number** (8 becomes 0) and put it as the **upper-left stick** on the **left number** (3 becomes 9).



Step 5. Read the digit. After fixing, the number on the right of the = sign is **0**.

Puzzle 4 digit = 0

Practice 5: Logic

Idea: there are only 4 cards, so there are 24 different ways to give them to Bilal, Hira, Mahnoor, Zoya. Each clue removes the ways that break it. We keep going until only one way is left.

Clue A (sum). "Bilal and Mahnoor hold cards that add up to 13."
Which two of the cards 4, 6, 7, 8 add up to 13? 6 + 7 = 13. So Bilal and Mahnoor must hold those two cards, in either order. This leaves 4 ways (out of 24):

Way	Bilal	Hira	Mahnoor	Zoya
1	6	4	**7**	8
2	6	8	**7**	4
3	7	4	**6**	8
4	7	8	**6**	4

Clue B (difference). "Bilal holds a card that is 1 more than the card held by Mahnoor."
This means Bilal's card minus Mahnoor's card = 1. Check every way above and cross out the ones where this is not true. This leaves 2 ways:

Way	Bilal	Hira	Mahnoor	Zoya
1	7	4	**6**	8
2	7	8	**6**	4

Clue C (bigger). "Mahnoor holds a bigger card than Zoya."
Check the ways above: Mahnoor's card must be bigger than Zoya's card. Only **one way** is left:

Way	Bilal	Hira	Mahnoor	Zoya
1	7	8	**6**	4

Read the answer. The question is: Which card is **Mahnoor** holding? In the last table, Mahnoor holds **6**.

Puzzle 5 digit = 6

Answers to the more-practice puzzles

Puzzle	Answer digit	Short working
Cipher (TOTK, shift 6)	9	Move each letter back 6 places. The word is NINE, so the digit is 9.
Pattern (3, 5, 10, 18, 29, ?)	3	The gaps are 2, 5, 8, 11. They go up by 3 each time, so the next gap is 14. The next number is 29 + 14 = 43. The last digit is 3.
Riddle	4	Cross out numbers clue by clue. Only 4 is left.
Matchstick	8	The fixed equation is 6 + 2 = 8.
Logic	2	Bilal holds card 2.

Stuck on a method? Look again at the full working for Practice 1 to 5. Every puzzle type in Code Crackers uses the same method, with different numbers and names.